

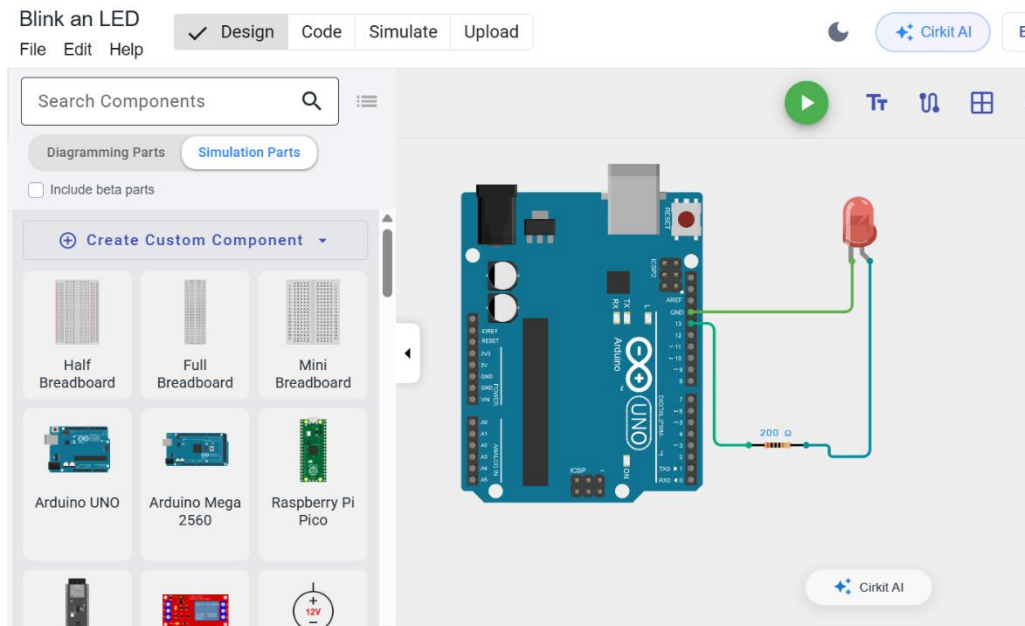
Experiments on Arduino Uno with Sensors

Yogendra Pratap Singh (19/09/2026)

Experiment-1 LED blinking & GPIO control.

(1.1) Blink an LED.

```
void setup() {  
  pinMode(13, OUTPUT);  
}  
  
void loop() {  
  digitalWrite(13, HIGH);  
  
  delay(1000);  
  
  digitalWrite(13, LOW);  
  
  delay(1000);  
}
```



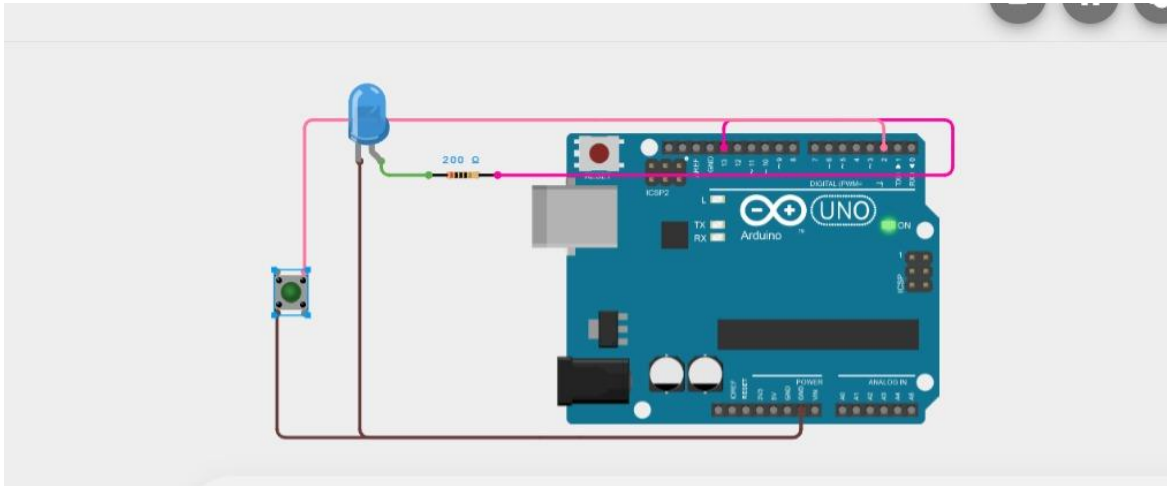
(1.2) Blink an LED and control it using push button.

```
const int buttonPin=2;  
const int ledPin=13;  
  
void setup() {  
  pinMode (ledPin, OUTPUT);  
  pinMode (buttonPin,INPUT_PULLUP);  
}
```

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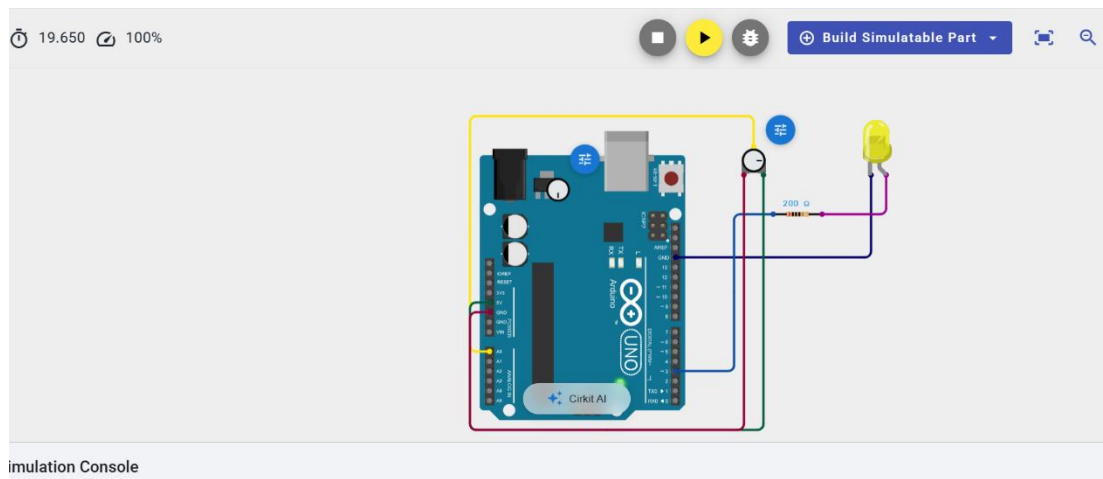
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```
void loop() {  
  int buttonState=digitalRead(buttonPin);  
  if (buttonState==LOW){  
    digitalWrite(ledPin,HIGH);  
  } else {  
    digitalWrite(ledPin,LOW);  
  }  
}
```



(1.3) Increase or decrease the LED illumination with Potentiometer.

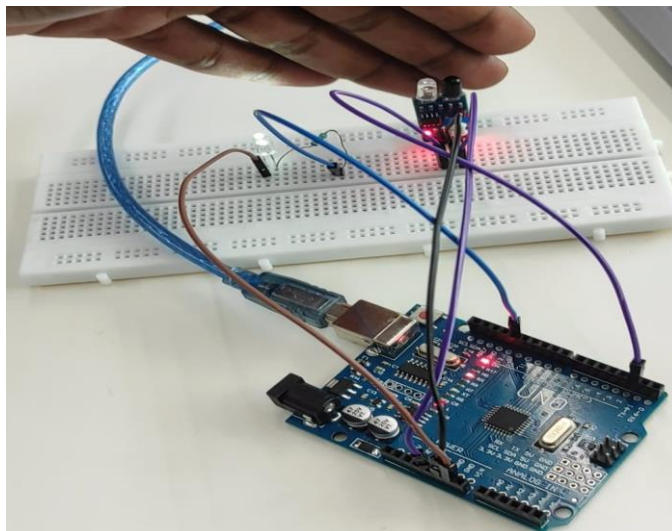
```
const int analogINPUT = A0;  
const int LED = 3;  
int inputVal = 0;  
void setup() {  
  pinMode (LED, OUTPUT);  
}  
void loop() {  
  inputVal = analogRead(analogINPUT);  
  analogWrite (LED, inputVal/4);  
  delay(100);  
}
```



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Experiment-02 Detect an object and display it on the Serial Monitor.

```
const int irSensorPin = 7;  
  
const int ledPin = 8;  
  
void setup() {  
  Serial.begin(9600);  
  
  pinMode(irSensorPin, INPUT);  
  pinMode(ledPin, OUTPUT);  
}  
  
void loop() {  
  int sensorValue = digitalRead(irSensorPin);  
  if (sensorValue == LOW) {  
    digitalWrite(ledPin, HIGH);  
    Serial.println("Object Detected!");  
  } else {  
    digitalWrite(ledPin, LOW);  
    Serial.println("No Object");  
  }  
  delay(500);  
}
```



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Experiment-03 To measure the distance of an object and display the distance on the Serial Monitor (Ultrasonic Distance Measurement).

```
const int trigPin=9;

const int echoPin=10;

long duration;

int distance;

void setup() {

pinMode(trigPin, OUTPUT);

pinMode(echoPin, INPUT);

Serial.begin(9600);

}

void loop() {

digitalWrite(trigPin, LOW);

delayMicroseconds(2);

digitalWrite(trigPin, HIGH);

delayMicroseconds(10);

digitalWrite(trigPin, LOW);

duration = pulseIn(echoPin, HIGH);

distance = duration * 0.034 / 2;

Serial.print("Distance: ");

Serial.print(distance);

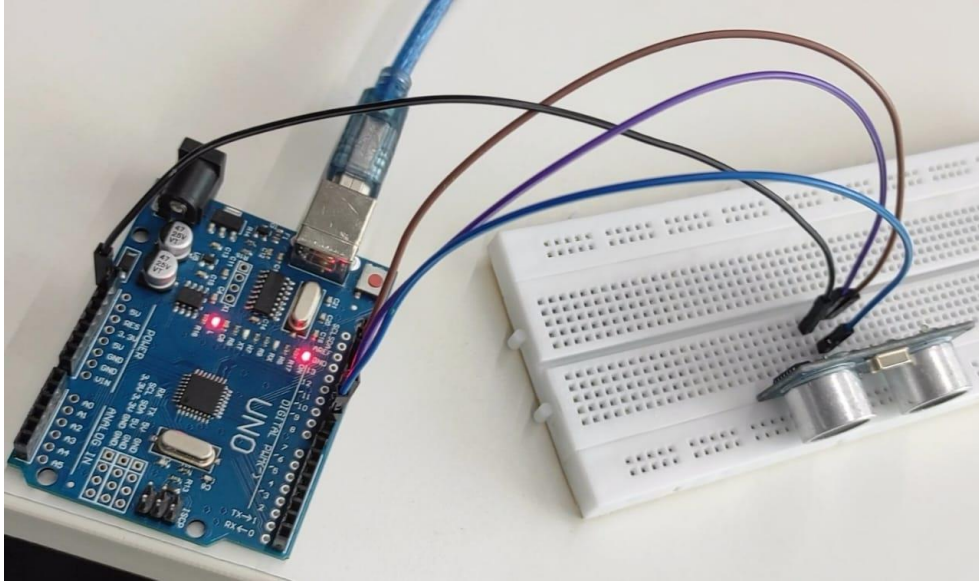
Serial.println("cm");

delay(200);

}
```

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Experiment-04 Connect a 0.96" OLED, any sensor and display sensor readings and messages.

```
#include <Wire.h>

#include <DHT.h>

#include <Adafruit_GFX.h>

#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128

#define SCREEN_HEIGHT 64

#define OLED_RESET -1

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT,
&Wire, OLED_RESET);

#define DHTPIN A0

#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

void setup() {

Serial.begin(9600);

dht.begin();
```

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```
if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {  
  Serial.println(F("SSD1306 allocation failed"));  
  for(;;);  
}  
  
display.clearDisplay();  
display.setTextColor(WHITE);  
display.setTextSize(2);  
display.setCursor(15, 24);  
display.print("Hello World!");  
display.display();  
}  
  
void loop() {  
}
```

